

Nicholas Mc-Donnell

R&D Software Engineer -- EDA / Computational Geometry

Santiago, Chile

namcdonnell@uc.cl +56 9 62061387

 GitHub: N9199  LinkedIn

Experience

Synopsys

Santiago, Chile

Senior R&D Engineer

April 2023 -- August 2026

- Migrated the entire C++ MDP codebase and all its dependencies to the next QSC (Qualified System Configuration) and C++ version.
- Fixing urgent production issues for several Photomask sites.
- Contributed to the implementation of Bézier-curve support for customer workflows.
- Implemented a debugging layer for the whole MDP software to be able to instantly dump and reproduce geometrical issues without having to send additional modified builds to the customers. This greatly reduced the debugging and fixing time from R&D and AEs.
- Significantly improved DP fracture task-scheduling performance for complex customer workloads.

Personal Projects

Ray Tracer — Rust

- Implemented a physically based ray tracer following *Ray Tracing in One Weekend*, extending it with an experimental BVH acceleration structure and multithreaded rendering.
- Restructured the rendering pipeline to enable tail-call optimization and improve the architecture for future performance work.
- Currently exploring Rust based GPU-accelerated rendering.

Segment Tree Library — Rust

- Designed and implemented a generic, high-performance segment tree library supporting range queries, range updates, and lazy propagation.
- Designed the API for flexibility across different segment-tree use cases while maintaining efficient implementations.
- Currently implementing wide-tree layouts to improve cache locality and performance.

Teaching and Research

Millenium Institute Foundational Research on Data

Santiago, Chile

Software Engineer

September 2020 -- December 2022

- Worked in projects about machine learning and interpretability.

Computer Science Department (PUC)

Santiago, Chile

Teacher Assistant

March 2021 -- December 2022

- Designed homework assignments and supplementary problems emphasizing algorithmic problem solving and non-obvious applications of techniques introduced in class.

Courses:

- Cryptography
- Algorithms & Data Structures
- Algorithm Design & Analysis

Mathematics Faculty (PUC)

Santiago, Chile

Teacher Assistant

March 2018 -- December 2020

- Led problem-solving sessions for undergraduate mathematics courses, working through exercises and guiding students through rigorous solution strategies in real time.

Courses:

- Real Analysis
- Abstract Algebra
- Several Calculus Courses

Education

Pontificia Universidad Católica de Chile (PUC)

Santiago, Chile

Mathematics B.Sc.

March 2017 -- December 2020

Coursework: Algorithms & Data Structures, Algorithm Design & Analysis, Probability & Statistics, Linear Algebra, Algorithms, Abstract Algebra

Activities: Model Theory Seminar Organizer, Mathematics Student Council, Mathematics Statutes Commission

Skills

Programming Languages: C++, Rust, Python, Java, SQL, Bash/Zsh

Tools: Linux, Git, Jujutsu, Vim, VSCode, Zed

Languages: Spanish (Native), English (professional/fluent)

Awards

International Collegiate Programming Contest Latin American Regional 2nd place in Chile (54th in Latin America)	2019
IEEEEXTREME 15.0 26th place	2021
IEEEEXTREME 16.0 23rd place	2022